



FACULTY OF COMPUTERS AND ARTIFICIAL INTELLIGENCE, CAIRO UNIVERSITY

IT443: Image Processing

Project: Face Recognition

Course Instructor:

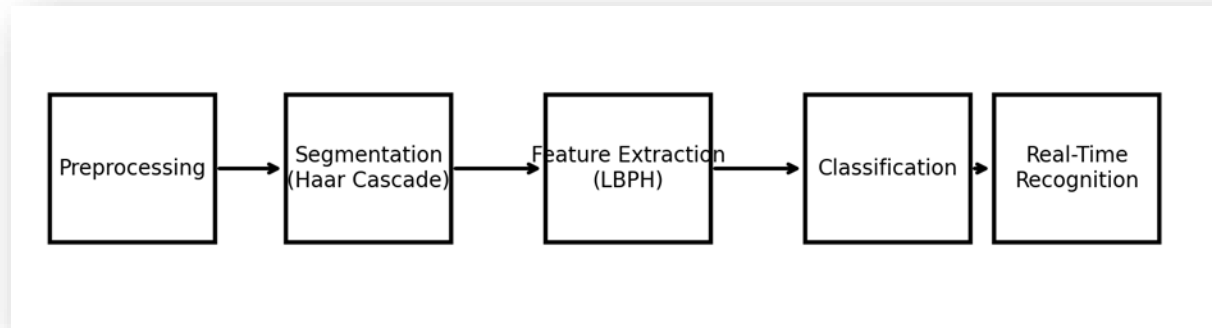
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Project Goal:

This project aims to design and implement a complete **Real-Time Face Recognition System** using Digital Image Processing techniques. The system satisfies the core course requirements by following a standard image processing pipeline that includes:



The application captures video frames from a webcam, detects and isolates human faces, enhances the facial images, extracts distinctive features, and identifies individuals in real time.

Pipeline:

1. Image Acquisition

Image Acquisition is the first stage of the proposed Face Recognition System. In this step, facial images are collected using a webcam connected to the computer. A custom Python script (`data_collector.py`) captures multiple images for each team member and dataset for different people under different conditions, including variations in facial expressions, head orientations, and lighting environments.

2. Preprocessing

After acquiring the facial images, a preprocessing stage is applied to enhance image quality and prepare the dataset for training. First, each image is converted to grayscale and the face region is detected using the Haar Cascade classifier. The largest detected face is cropped and resized to a fixed size of **200 × 200 pixels** to ensure consistency across all samples.

To improve image contrast and reduce the effect of uneven lighting, **Contrast Limited Adaptive Histogram Equalization (CLAHE)** is applied with a clip limit of 2.0 and a tile grid size of 8×8 . This technique enhances important facial details while preventing excessive amplification of noise.

To increase the size and diversity of the dataset, several data augmentation techniques are performed, including horizontal flipping, small rotations (-10° , -5° , $+5^\circ$, and $+10^\circ$), and brightness

adjustments. These transformations generate additional training samples that help the recognition model become more robust to changes in pose and illumination.

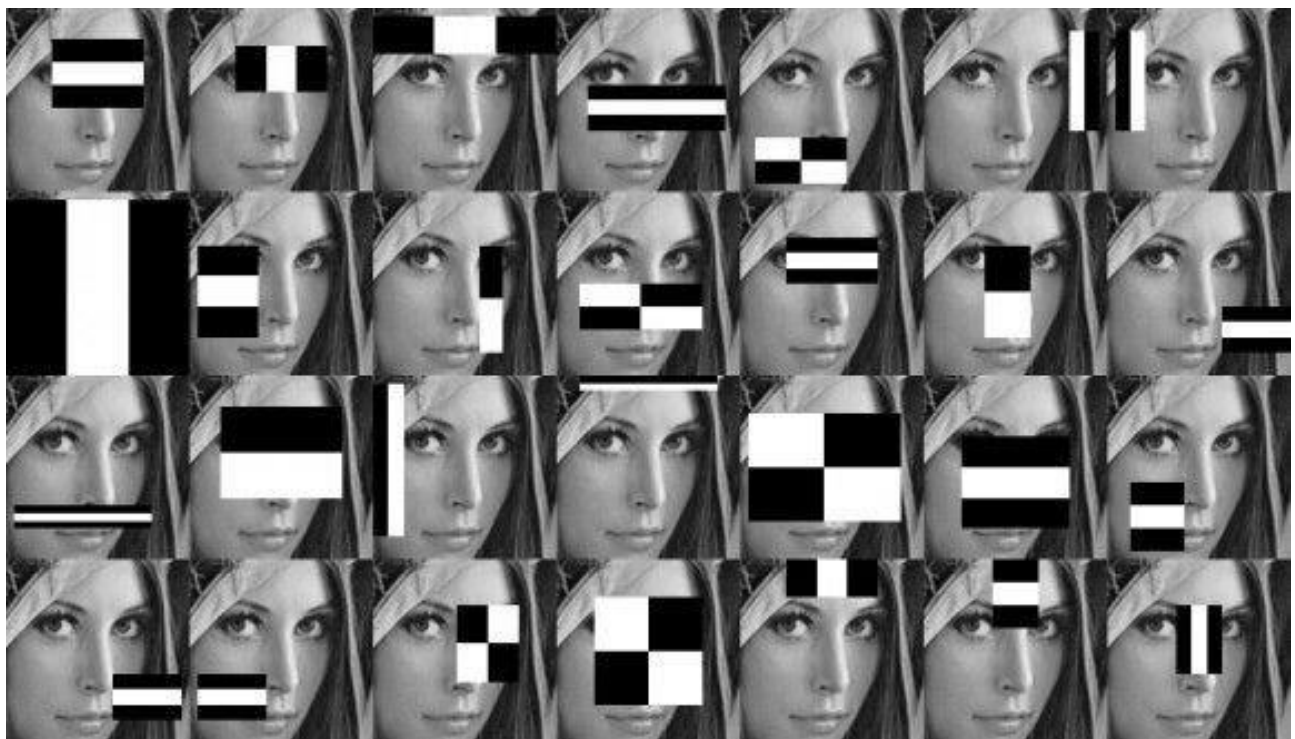
The resulting processed and augmented images are stored in the `processed_dataset/` directory and used as the final input for the feature extraction and training stages

3. Segmentation

The segmentation stage is implemented through face detection and region isolation using the **Haar Cascade classifier**. Although classical image segmentation typically refers to partitioning an image into meaningful regions, in face recognition the primary objective is to separate the facial area from the surrounding background.

The Haar Cascade algorithm scans each webcam frame and identifies rectangular regions that are likely to contain a human face. Once a face is detected, the corresponding bounding box is used to crop the facial region and discard irrelevant background information such as walls, clothing, and other objects. If multiple faces are detected, the largest face is selected as the region of interest.

This cropped face represents the segmented output of the system and is passed to the preprocessing and feature extraction stages. By isolating only the relevant facial region, the system reduces computational complexity and ensures that subsequent processing focuses exclusively on discriminative facial features.



4. Feature Extraction

In the feature extraction stage, the segmented facial image is transformed into a numerical representation that captures its unique texture characteristics. This project uses the **Local Binary Patterns Histograms (LBPH)** algorithm, which is widely used in face recognition due to its simplicity, efficiency, and robustness to moderate illumination changes.

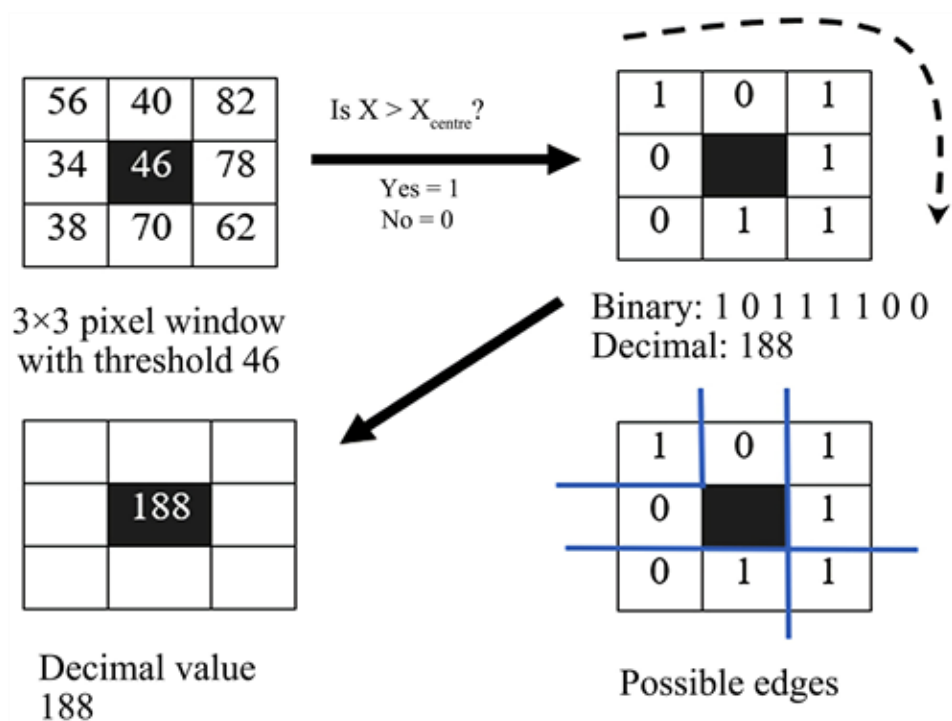
LBPH analyzes each pixel by comparing it with its surrounding neighboring pixels. For every neighbor, a binary value is assigned: **1** if the neighbor's intensity is greater than or equal to the center pixel, and **0** otherwise. The resulting binary pattern is converted into a decimal value known as the Local Binary Pattern (LBP) code.

The face image is then divided into several small regions, and a histogram of LBP codes is computed for each region. These histograms are concatenated to form a single feature vector that describes the local texture distribution of the entire face.

This feature vector serves as a compact and discriminative representation of the face and is used as the input to the classification stage for identity recognition.

LBPH Algorithm – Simple Example

Suppose the center pixel has intensity 100. Each neighboring pixel is compared with 100. Values greater than or equal to 100 are marked as 1, otherwise 0. For example, the binary pattern 11001011 corresponds to decimal value 203. This code describes the local texture around that pixel.



5. Classification

In the classification stage, the extracted feature vectors are used to identify the person by comparing them with previously learned data. The system is trained using the **LBPH recognizer** provided by OpenCV, where each face feature vector is associated with a specific label (person ID).

During training, the model stores the histogram representations of all registered individuals. In the recognition phase, the system extracts features from a new input face and computes the distance between this feature vector and all stored templates. The identity corresponding to the closest match (minimum distance) is selected as the predicted class.

The classifier also returns a confidence value, which represents how similar the detected face is to the trained samples. A lower confidence value indicates a better match, while higher values suggest uncertainty. Based on a predefined threshold, the system can either assign a known identity or label the face as “Unknown” if the confidence is not sufficient.

This approach allows the system to perform real-time face identification with relatively low computational cost and acceptable accuracy for small to medium-sized datasets

5. Real-Time Recognition

The system works in real time using a webcam feed, where each frame is processed continuously. Faces are detected using Haar Cascade, then passed through the same preprocessing steps to maintain consistency. After that, LBPH extracts features and the trained model classifies the identity.

If the confidence value is below a defined threshold, the person’s name is displayed on the screen; otherwise, the face is labeled as **Unknown**. This process runs continuously for every frame to enable live recognition.

Challenges and Limitations

During the development of the system, several challenges were encountered:

- **Lighting variations:** Strong or weak lighting affected detection and recognition accuracy.
- **Pose changes:** Large head rotations or side profiles reduced detection performance.
- **Similar faces:** People with similar facial features sometimes produced close confidence scores.
- **Small dataset size:** Limited training images reduced model generalization.
- **Haar Cascade limitations:** Detection may fail with occlusions or non-frontal faces.
- **Threshold tuning:** Choosing the correct confidence threshold for “Unknown” classification was difficult and affected accuracy.

These challenges were partially mitigated using preprocessing, augmentation, and parameter tuning, but they remain inherent limitations of classical face recognition systems.

Team Project Summary:

Person	Role	Deliverables
Hady Hassan	Data & Preprocessing	• <code>data_collector.py</code>, for capture images for each member in the team • <code>image_enhancer.py</code>, for removing noise and more advanced enhancement and data augmented
NourEldin Mohamed	Modelling & Training	• <code>trainer.py</code>, using LBPH algorithm for extract feature and training
Abdallah Kareem	System Integrator	• <code>face_recognizer.py</code>, integrate all parts with one file • Report Summary